CSA06507-Generative AI

Lab Experiment's

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Unit 2: Lab Programs for Prompting

Experiment 12: Prompt Evaluation, Comparison and Refinement Using LLM Responses

Aim:

To design multiple prompts for a given text-generation task, evaluate their generated outputs using predefined criteria such as relevance, accuracy, completeness, clarity, and format adherence, and refine the best-performing prompt to improve response quality.

Algorithm

1. Start the program.
2. Select a text-generation task.
3. Design multiple prompts for the same task.
4. Generate responses using each prompt.
5. Evaluate the responses based on relevance, accuracy, completeness, clarity, and format adherence.
6. Assign scores to each evaluation criterion.
7. Compare the total scores of all prompts.
8. Select the best-performing prompt.
9. Refine the selected prompt to improve its output.
10. Display the final refined response and evaluation results.
11. Stop the program.

Python Program

Google Colab Code:

Copy and run the following code in Google Colab.

Python

Run

```
# Prompt Evaluation, Comparison and Refinement


# Step 1: Define the task
task = "Explain the benefits of Artificial Intelligence in education."


# Step 2: Design multiple prompts
```

```
prompts = {
    "Prompt 1": "Explain Artificial Intelligence in education.",

    "Prompt 2": """Explain the benefits of Artificial Intelligence
    in education in simple English with examples.""",

    "Prompt 3": """Explain five benefits of Artificial Intelligence
    in education. Use headings, simple language and examples.
    Write a short conclusion."""
}

# Step 3: Generate sample responses
responses = {
    "Prompt 1": "AI helps students learn. It supports teachers and improves education.",

    "Prompt 2": """AI provides personalized learning.
    It helps students understand difficult topics.
    It also supports teachers in evaluating students.""",

    "Prompt 3": """1. Personalized Learning: AI provides learning based on student needs.
    2. Quick Feedback: AI gives immediate feedback.
    3. Smart Tutoring: AI explains difficult concepts.
    4. Saves Time: AI helps teachers automate tasks.
    5. Better Assessment: AI supports student performance analysis.

    Conclusion: AI makes education more effective and accessible."""
}

# Step 4: Define evaluation scores
# Scores are given out of 5
```

Input

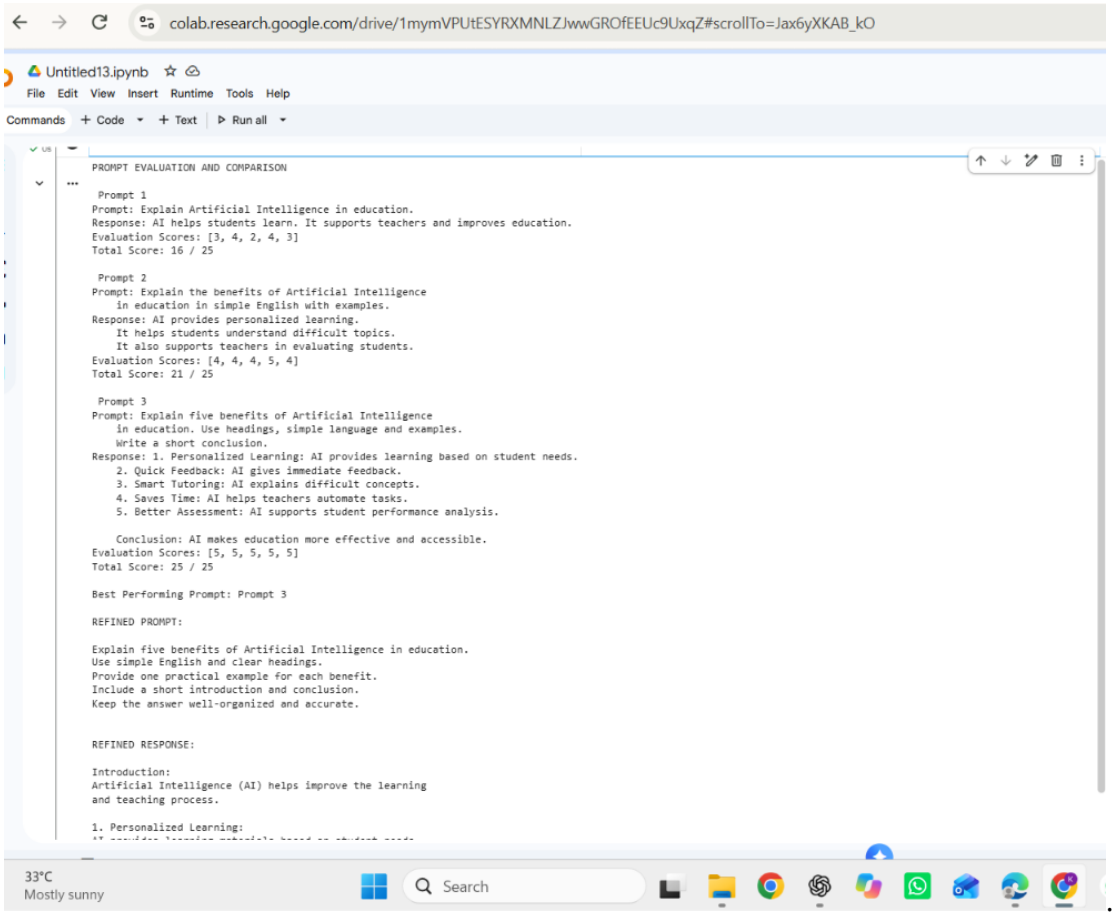
Text-generation task:

Plain text

Explain the benefits of Artificial Intelligence in education.

Three different prompts are designed for the same task and evaluated.

Output



Evaluation Criteria

Criteria	**Description**
Relevance	Checks whether the response matches the task.
Accuracy	Checks whether the information is correct.
Completeness	Checks whether all required points are included.
Clarity	Checks whether the response is easy to understand.
Format Adherence	Checks whether the required format is followed.

Result

Thus, prompt evaluation, comparison, and refinement using LLM responses were successfully demonstrated in Google Colab. Three different prompts were designed and evaluated using five predefined criteria. The best-performing prompt was selected and refined to generate a more relevant, accurate, complete, clear, and well-structured response.